Revision

COMP1048: Databases and Interfaces (2025-2026)

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Overview

This Lecture

- Exam format and content
- Module content
- Questions

Exam Format and Content

Exam Format

- Answer all Three questions:
- Q1 - Web Programming [15 Marks]
- Q2 - Relational DB Design, Theory and **DBMS** [15 Marks]
- Q3 - **SQL** and **ER** Design [20 Marks]
- Worth 50% of the final mark
- Time: 2 hours

Exam Content

- Anything covered in the course may come up in the exam:
  - `DBMS`
  - Relational model and algebra
  - `ER Diagram`
  - Normalization
  - `SQL` Queries
  - `HTML` and `CSS`
  - `Flask`
  - ...etc.
- Mock exam questions
- Past exam papers

Module Content

Database: Relational Model

- What is Database? Why we need it?
- What is `DBMS`?
- Relational model
  - Relations, attributes, domains, relationship, cardinality ratio
  - Candidate keys, primary keys and foreign keys
  - Referential integrity
- Relational algebra
  - Operators: `selection`, `projection`, `set operators`, `join operators` and `rename`
  - Write relational algebra based on the problem description

Database: Database Design

- ER Diagram
  - Entity
  - Relationship
  - Attributes
  - Cardinality
  - Draw ER diagram according to the problem description
- Normalization
  - What is normalization? why we need it?
  - What is the functional dependency?
  - What are `1NF`, `2NF` and `3NF`?
  - How to convert unnormalized data into `3NF`?

Database: SQL Queries

- What is SQL?
- How to Create and Delete a table?
- How to Insert and Select data?
- How to Update and Delete data?
- Alias, subqueries, aggregate functions, grouping, joins
- What's the connection and differences between SQL and Relational Algebra?

Interfaces: HTML and CSS

- What is HTTP?
- What is HTML?
- Syntax of HTML
  - Images
  - Hypertext links
  - Lists
  - Tables
  - Forms
- Syntax of CSS
- How to add CSS to HTML
- What is DOM?

Interfaces: Web Application

- Python
- Flask
- Jinja
- The client-server model
- What are Web applications?
- Why we need web applications?
- Java-scripts and client-side scripting (understand only)

How to Prepare

Approaches

- Text book
- Lecture slides
- Recordings
- Lab excercises
- Mock exam questions and Past exam paper

Common Errors in SQL Queries (from Quiz)

- Syntax Error: e.g., Missing comma(`,`), semicolon(`;`), Use `WHERE` together with `GROUP BY`, ...
- Incorrect ordering: Lacked `ORDER BY`, `ORDER BY` on the wrong column, or `ORDER BY` with wrong order.
  - A notable problem: `ORDER BY` a string, e.g., `ORDER BY "name"`
- `IS NULL`: Use of `= NULL`, `= "NULL"` or `=` instead of `IS NULL`
- `NATURAL JOIN`: Many solutions made use of "`NATURAL JOIN` when two tables have no common attributes
- `CROSS JOIN`: Misused `CROSS JOIN` resulting in nonsensical results.
- `LEFT JOIN`: Incorrect use of `LEFT JOIN` resulted in additional (incorrect) values being returned.

Common Errors (cont)

- `INNER JOIN`: Many solutions that made use of `INNER JOIN` with incorrect conditions.
- Ambiguous names: `SQL` statement errors arising from columns with the same name in different tables (during a `JOIN`).
- Commonly seen when using `ORDER BY`.
- Aggregate function: Missing corresponding `GROUP BY` clause when using `aggregate functions`, or incorrect `aggregate function` were used, e.g., use `SUM` to calcualte the average.
- Duplicated Values: Duplicate tuples/rows in the resulting table.